

New perspectives of curcumin in cancer prevention

.

Numerous natural compounds have been extensively investigated for their potential for cancer prevention over decades. Curcumin, from *Curcuma longa*, is a highly promising natural compound that can be potentially used for chemoprevention of multiple cancers. Curcumin modulates multiple molecular pathways involved in the lengthy carcinogenesis process to exert its chemopreventive effects through several mechanisms: promoting apoptosis, inhibiting survival signals, scavenging reactive oxidative species (ROS), and reducing the inflammatory cancer microenvironment. Curcumin fulfills the characteristics for an ideal chemopreventive agent with its low toxicity, affordability, and easy accessibility. Nevertheless, the clinical application of curcumin is currently compromised by its poor bioavailability. Here we review the potential of curcumin in cancer prevention, its molecular targets, and action mechanisms. Finally, we suggest specific recommendations to improve its efficacy and bioavailability for clinical applications.